Automation in Construction 107 (2019) 102919

Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon

Review

Mapping computer vision research in construction: Developments, knowledge gaps and implications for research

Botao Zhong[a], Haitao Wu[a,*], Lieyun Ding[a], Peter E.D. Love[b], Heng Li[c], Hanbin Luo[a], Li Jiao[a]

[a] School of Civil Engineering & Mechanics, Huazhong University of Science & Technology, Wuhan, 430074, China
[b] School of Civil and Mechanical Engineering, Curtin University, GPO Box U1987, Perth, WA 6845, Australia
[c] Department of Building and Real Estate, The Hong Kong Polytechnic University, Kowloon, Hong Kong, China

ARTICLE INFO

Keywords:
Computer vision
Construction
Science mapping
Review

ABSTRACT

Computer vision is transforming processes associated with the engineering and management of construction projects. It can enable the acquisition, processing, analysis of digital images, and the extraction of high-dimensional data from the real world to produce information to improve managerial decision-making. To acquire an understanding of the developments and applications of computer vision research within the field of construction, we performed a detailed bibliometric and scientometric analysis of the normative literature from 2000 to 2018. We identified the primary areas where computer vision has been applied, including defect inspection, safety monitoring, and performance analysis. By performing a mapping exercise, a detailed analysis of the computer vision literature enables the identification of gaps in knowledge, which provides a platform to support future research in this fertile area for construction.

1. Introduction

Computer vision is an interdisciplinary scientific field that focuses on how computers can acquire a high-level understanding from digital images or videos [1]. It can be used to transform the tasks of engineering and management in construction by enabling the acquisition, processing, analysis of digital images, and the extraction of high-dimensional data from the real world to produce information to improve decision-making. Furthermore, computer vision can provide practitioners with rich digital images and videos (e.g., location and behavior of objects/entities, and site conditions) about a project's prevailing environment and therefore enable them to better manage the construction process [2].

Computer vision has been used to examine specific issues in construction such as tracking people's movement [3,4], progress monitoring [5], productivity analysis [6], health and safety monitoring [7,8], and postural ergonomic assessment [9]. In bringing together the developments and applications of computer vision research, we reviewed the normative literature to identify emerging trends to provide a roadmap for future lines of inquiry. We acknowledge that reviews of computer vision have been undertaken for specific problem domains such as defect detection and condition assessment [10] and safety [2], but they are subjective and therefore are prone to bias [11].

To address such limitations, we conducted a science mapping analysis of the computer vision literature from 2000 to 2018 [12–14]. Our aim is to garner an understanding of the intellectual core of the computer vision domains in construction instead of focusing on individual and specific issues. In pursuing this line of inquiry, we unearthed the primary areas and topics of research that computer vision has focused upon, the gaps that restrict its application, the individuals and institutions that are the forefront of its developments, the collaborations that take place, and the primary publication outlets. We believe the line of inquiry presented in this paper will not only advance our understanding of computer vision research in construction and engineering but enhance the field's ability to explore how technological innovation can be effectively applied to address problems confronting everyday practice.

The paper commenced by introducing and describing the research method we have adopted to conduct our review (Section 2). We then discussed the results of our bibliometric and scientometric analysis (Section 3). Next, we categorized the areas of computer vision research and identified knowledge gaps that exist in the literature (Section 4). In the final section of our paper, we presented our conclusions and limitations.

2. Research method

Our bibliometric review and scientometric analysis of the computer

* Corresponding author.
E-mail address: haitao_w@hust.edu.cn (H. Wu).

https://doi.org/10.1016/j.autcon.2019.102919
Received 4 April 2019; Received in revised form 21 July 2019; Accepted 22 July 2019
Available online 30 July 2019

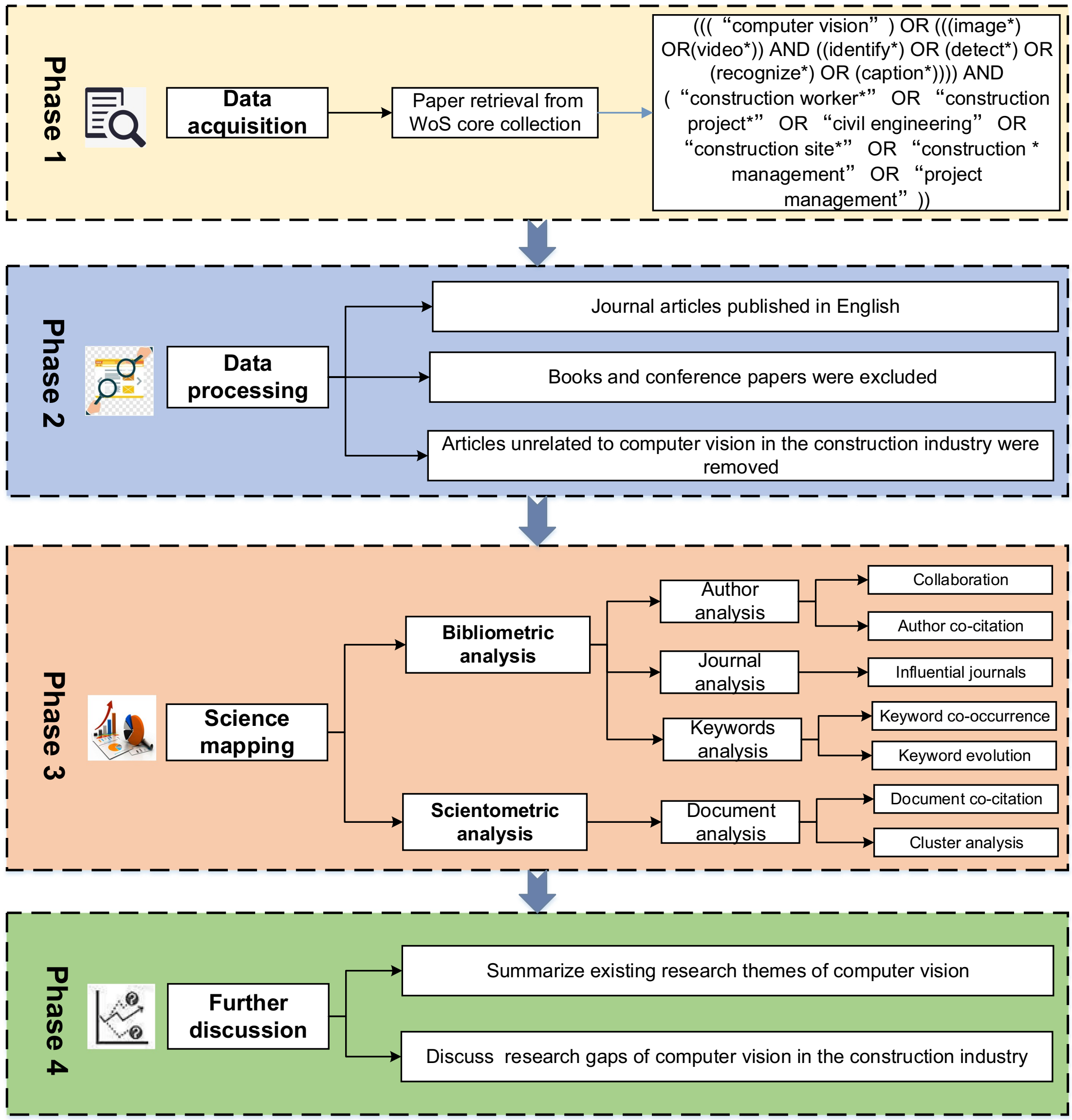


Fig. 1. The outline of the research design.

vision literature utilized the Web of Science (WoS) database. We were drawn to the WoS as it contains the most comprehensive and influential journals within the field of construction and engineering management [13,15]. In this paper, the academic relationships and keywords within the domain of computer vision in construction are mapped, the salient research themes identified using cluster analysis, which are then further explored by utilizing a qualitative line of inquiry. Additionally, we discussed the gaps in knowledge that exist within the computer vision literature. We presented an overview of the research process in Fig. 1.

2.1. Data acquisition

We applied the following search string within the WoS data to identify papers for our review that commences in 2000 and finishes in 2018: (((“computer vision”) OR (((image*) OR(video*)) AND ((identify*) OR (detect*) OR (recognize*) OR (caption*)))) AND (“construction worker*” OR “construction project*” OR “civil engineering” OR “construction site*” OR “construction * management” OR “project management”)), where “*” denotes a fuzzy search. We identified a paper when the defined terms appeared in its title, keywords, or abstracts. We also restricted our search to articles in peer-reviewed scholarly journals, which had been published in English as they are reputable and reliable sources [16]. A manual review on search results was adopted to remove unrelated papers. We hasten to note that the exclusion of articles from books, editorials, and conference papers. As a result of conducting our search, we identified a total of 216 journal articles were suitable for analysis.

2.2. Bibliometric and scientometric analysis

After the acquisition and processing of the journal articles, the next step was to perform a quantitative analysis using bibliometric and scientometric techniques. The bibliometric analysis aimed to scientifically map and visualize the dataset [17] by examining themes such as authors, journals, and keywords. Contrastingly, scientometric techniques were used to quantitatively analyze and assess the content of publications [13]. There are two general scientometric approaches: (1) normative; and (2) descriptive [18].

The normative approach aims to create norms, rules, and heuristics to ensure progress within a particular field. Equally, the purpose of the descriptive approach is to observe and report on the actual activities within a field, and in particular, refer to its leading researchers'. We have adopted a descriptive approach, as according to Serenko et al. [19], "it fits better with a quantitative analysis of scientific publications" (p.5). Noteworthy, the difference between the normative and descriptive approaches can be unclear at times. For example, a "quantitative analysis of collaboration patterns of leading researchers may lead to the development of normative recommendations" [19]. CiteSpace, however, can enable knowledge domains to be systematically created using an array of graphs to visualize and analyze the literature [20]. Moreover, the developments and research progress and its corresponding time nodes and emergent trends were able to be identified [12].

Two quantitative metrics to determine the critical nodes in a network were: (1) the betweenness centrality; and (2) burst strength. A node with high betweenness centrality indicates there is a degree of control over the flow of information across a network, which therefore identifies the potential for boundary spanning and transformative discoveries [21]. In a network, there are two types of nodes that may have high betweenness centrality scores. These nodes are those that are [22]: (1) highly connected to others; and (2) positioned between different groups. The centrality is calculated using Eq. (1), where the P_{jk} represents the number of shortest paths between $node_j$ and $node_{ik}$, and the $P_{jk}(i)$ represents the number of paths that pass through $node_i$ [23]. Noteworthy, CiteSpace uses Kleinberg's algorithm [24] to detect citation bursts, which provides an indicator of the most active areas of research. Such burst can last a single year or may occur over an extensive period of time.

$$\text{Centrality}(node_i) = \sum_{i \neq j \neq k} \frac{P_{jk}(i)}{P_{jk}} \tag{1}$$

In our study, we analyzed the following: (1) authors; (2) journals; (3) keywords; and (4) documents and clusters. This level of analysis was in line with other studies that have performed a scientometric analysis in construction [11]. The analysis of authors included determining their citations and collaborations with other scholars and institutions. Our analysis aimed to determine the type and volume of publications and the most influential journals. In the case of keyword analysis, we aimed to identify emerging topics of interest in this domain by determining the co-occurrences and the network's evolution. In the final stage of our analysis, we utilized document analysis to identify high cited papers, and cluster analysis to classify the cited documents to provide the basis to identify emerging research themes.

3. Science mapping

After data processing, a series of networks were generated to determine the state of play for computer vision research. The 216 journal articles were listed according to their year of publication. Then, networks of authors (collaboration and author co-citation), journals, keywords, documents, and clusters were derived using science mapping.

3.1. An overview of the sampled literature

Fig. 2 presented a distribution of research articles published within the area of computer vision in the construction industry for over 18 years. Before 2010, there had been limited research though most tended to focus on object recognition [25–27]. With the emergence of deep learning, which simplifies the process of feature extraction using convolution, we anticipate that computer vision research will become a significant area of research and as a result, we expect a proliferation of scholarly works in this area in the near future.

3.2. Author analysis

The author analysis focused on collaboration and co-citation between authors. We defined research collaboration as working together

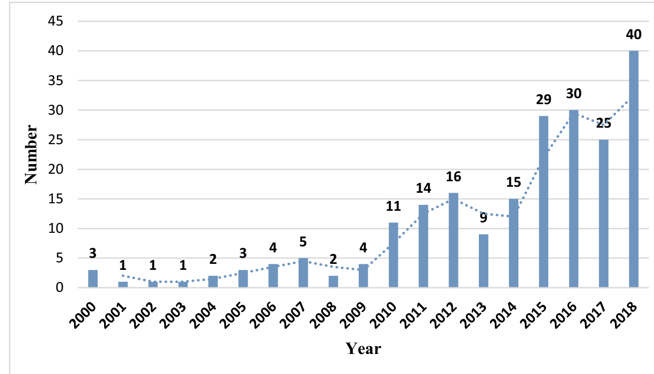


Fig. 2. Number of computer vision articles in WoS from 2000 to 2018.

to achieve the common goal of producing new scientific knowledge [28]. Collaboration analysis in this study was undertaken to illustrate relationships from a micro to a macro perspective; that is, between scholars and institutions. The identification of the existing collaborations within a specific domain can improve access to expertise and funding opportunities and provide a platform for sharing and exchanges ideas and findings [11]. The author co-citation analysis identified the relationships between authors cited in the same publication, which offered an approach for understanding the intellectual structure of computer vision research [29].

3.2.1. Collaboration

In Fig. 3, we exhibited a co-authorship network in which there are a total of 50 nodes and 70 links. The links between the nodes signify the degree of collaboration that prevails. Table 1 showed the top ten nodes with high levels of collaboration frequency.

Three research communities have formed a sphere of influence centered around a figurehead. For example, we can see that Brilakis Ioannis K was the principal author within his research community, which included Dai Fei and Park Manwoo. Similarly, Ding Lieyun was the primary author of a research community that consisted of Luo Hanbin, Love Peter E. D., Zhong Botao, Fang Weili, and Fang Qi. Similarly, Li Heng has acted as the figurehead within his research community, which incorporated an array of scholars including Luo Xiaochun, Huang Ting, Cao Dongping, and Yang Xincong.

These three primary authors possess high scores in terms of their betweenness centrality were: (1) Brilakis Ioannis K. (centrality = 0.05); (2) Ding Lieyun (centrality = 0.05); and (3) Li Heng (centrality = 0.04). The aforementioned scholars have made a significant impact on developing and engendering collaboration within the construction and engineering community within the domain of computer vision. Several productive authors, however, such as Kim Hyoungkwan (frequency = 6) has had limited international collaboration with others in the field. We suggest that collaboration is pivotal to extending and developing computer vision in construction, mainly to ensure that research has relevance to practice.

In Fig. 4, we showed a network revealing the collaborations between institutions comprising 33 nodes and 24 links. This analysis aimed to identify those institutions that have made a significant contribution to the field of computer vision in construction. The size of nodes indicates the number of published articles per institution, while the link highlights the collaboration between two institutions.

The most productive institutions were: Georgia Institute of Technology (14 published articles), Hong Kong Polytechnic University (10 published articles), University of Illinois (9 published articles), Yonsei University (9 published articles), University of Michigan (8 published articles) and Huazhong University of Science and Technology (8 published articles).

The betweenness centrality was most influential with the University

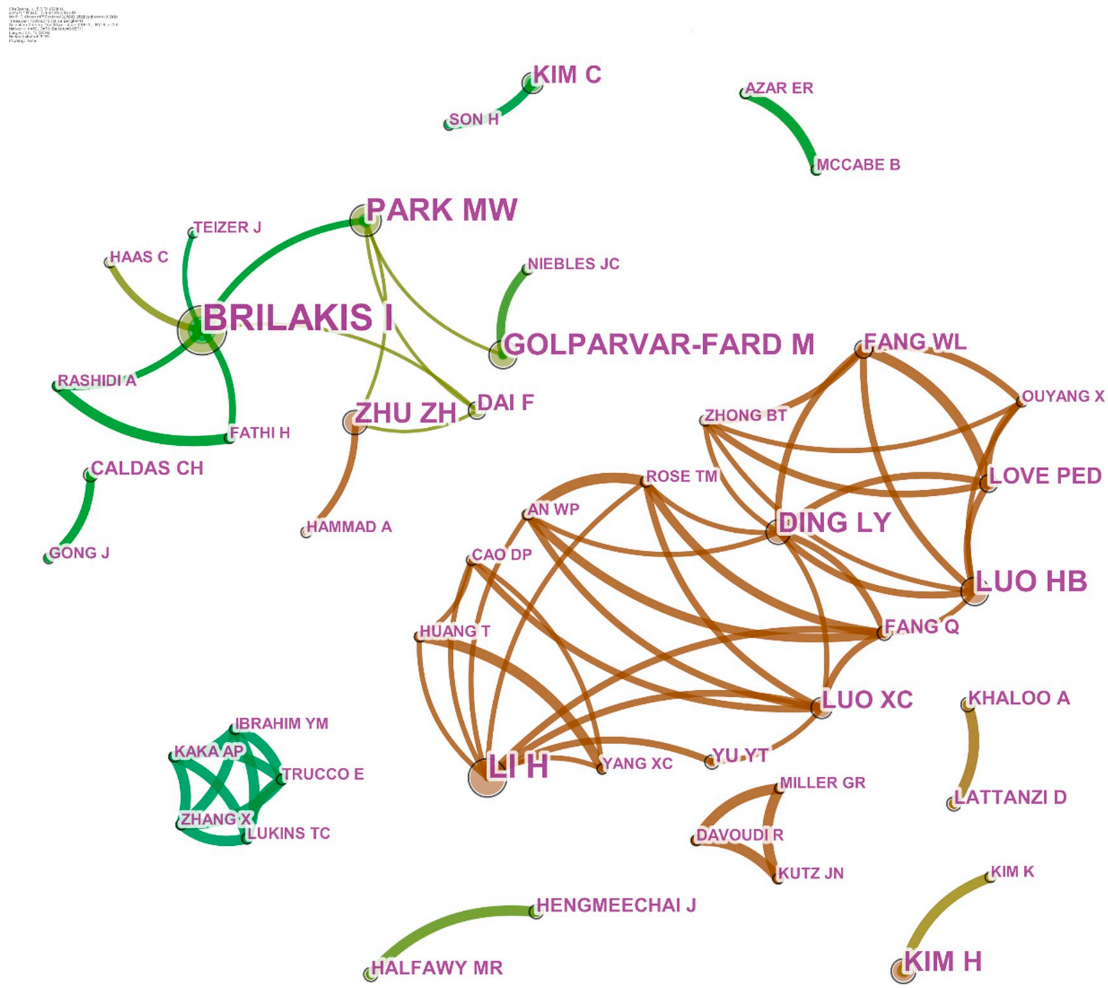


Fig. 3. The collaboration network of productive authors.

of Michigan (centrality = 0.05), followed by the University of Illinois (centrality = 0.46), then the Georgia Institute of Technology (centrality = 0.34), Tongji University (centrality = 0.34) and finally the Hong Kong Polytechnic University (centrality = 0.33). However, the betweenness centrality was high, which suggested that these universities engaged in promoting academic exchange and the sharing of intellectual knowledge.

Two institutional clusters were identified in Fig. 4 involving: (1) Georgia Institute of Technology, University of Illinois, Myongji University and University of Michigan; and (2) Hong Kong Polytechnic University, Huazhong University of Science and Technology and Curtin University. Among these institutions, the University of Michigan has acted as a critical node in the academic collaboration between institutional clusters.

If we examine the citation burst, then Georgia Institute of Technology (burst strength = 5.2373, 2010–2012) and Hong Kong Polytechnic University (burst strength = 3.5965, 2017–2019) were at the forefront of making an impact with their research. Of note, the outputs of Hong Kong Polytechnic University have only emerged in the last four years, but the quality and impact of this work has received worldwide attention. With this in mind, there is a likelihood that Hong Kong Polytechnic University under the research leadership of Li, Heng will be central to further developing the field in the future.

3.2.2. Author co-citation

The author co-citation analysis identifies the relationships among

Table 1
Top ten collaboration nodes.

Author[a]	Institution	Country	Frequency
Brilakis Ioannis K	University of Cambridge	United Kingdom (UK)	19
Li Heng	Hong Kong Polytechnic University	China	11
Golparvar-Fard Mani	The University of Illinois at Urbana-Champaign	United States of America (USA)	10
Zhu Zhenhua	Universite Concordia	Canada	10
Park Manwoo	Myongji University	South Korea	9
Kim Hyoungkwan	Yonsei University	South Korea	9
Luo Hanbin	Huazhong University of Science and Technology	China	7
Ding Lieyun	Huazhong University of Science and Technology	China	6
Dai Fei	West Virginia University	USA	6
Luo Xiaochun	Hong Kong Polytechnic University	China	5

[a] Authors surname is presented first throughout our paper.

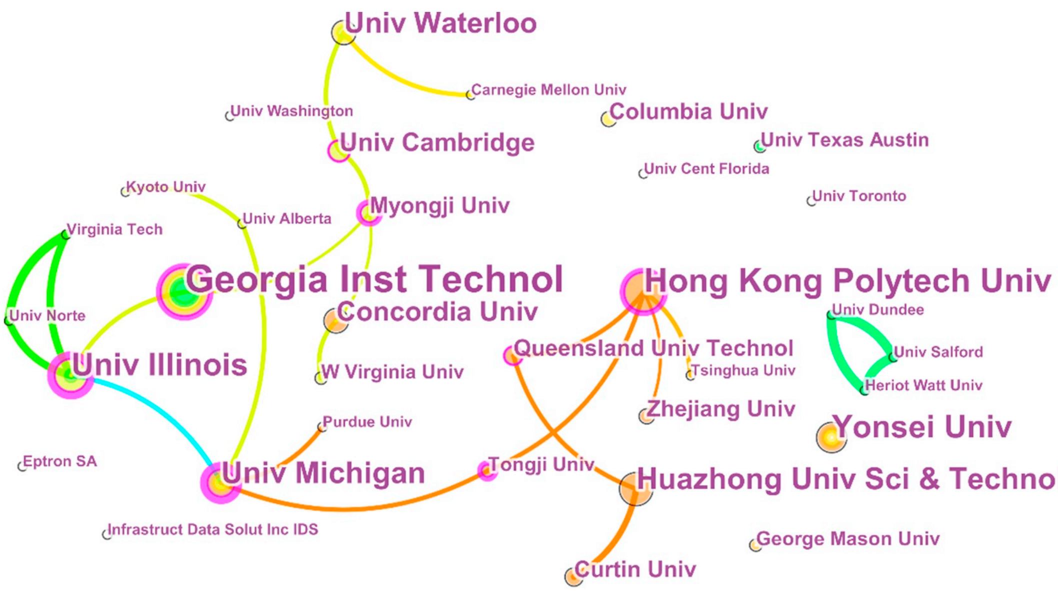


Fig. 4. The collaboration network of productive institutions.

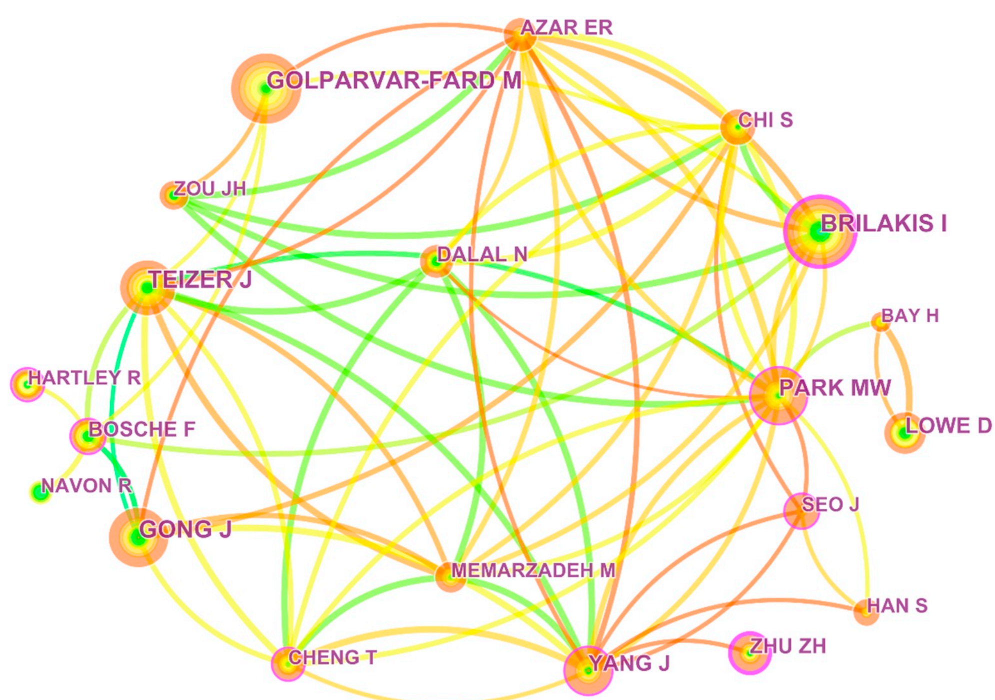


Fig. 5. Author co-citation map.

those authors that are cited in the same publication. Setting the minimum number of co-citations at 20 in CiteSpace, a total of 20 out of the 334 nodes met the thresholds. In Fig. 5, the node size expresses the number of co-citations for each author. The links between authors reflect the established indirect cooperative relationships based on citation frequency.

Table 2 identified the top ten cited authors. The top ten most highly cited authors were from USA (3); UK (2); Canada (2); South Korea (2); China (1); and German (1). Of these authors, Golparvar-Fard Mani, the director of real-time and automated monitoring and control laboratory at the University of Illinois at Urbana-Champaign, has focused on developing scientific solutions for automated and real-time performance monitoring. Conversely, Brilakis Ioannis K. has specialized in the field of automation and used computer vision to examine areas such as

Table 2
Top 10 cited authors.

Author	Frequency	Country	Centrality
Golparvar-Fard Mani	60	USA	0.09
Brilakis Ioannis K	57	UK	0.22
Gong Jie	52	USA	0.05
Teizer Jochen	51	Germany	0.07
Park Manwoo	48	South Korea	0.12
Lowe David	39	Canada	0.04
Yang Jun	38	China	0.15
Zhu Zhenhua	33	Canada	0.24
Chi Seok-ho	32	South Korea	0.08
Bosché Frederic	31	UK	0.19

Table 3
Main journal outlets.

Journal	Country	Count	Percentage
Automation in Construction	Netherlands	67	31%
ASCE Journal of Computing in Civil Engineering	USA	35	16%
Advanced Engineering Informatics	UK	33	15%
Computer-aided Civil and Infrastructure Engineering	USA	7	3%
Structural Control & Health Monitoring	USA	7	3%

multimedia data analysis, classification, retrieval, and processing and the three dimensional (3D) reconstruction.

As shown in Table 2, a highly co-cited author may not have a high betweenness centrality. However, when a node has a high citation frequency, and centrality, it indicates the author has had a fundamental influence on the development of computer vision research. These authors were: (1) Brilakis Ioannis K. (frequency = 61, centrality = 0.22); and (2) Zhu Zhenhua (frequency = 46, centrality = 0.24). In the co-citation, several computer scientists were also identified such as Lowe David who presents an algorithm that applies a priority search on hierarchical k-means trees to solve nearest neighbor matching in high-dimensional spaces [30].

3.3. Journals analysis

The identification of journals that have published leading scholarly works is an important source for researchers aiming to acquire insights into the latest developments and emerging trends with a field [11]. With this in mind, we analyzed the 216 articles to determine the main research outlets for computer vision research. Table 3 indicates that *Automation in Construction* is the leading scholarly journal for computer-vision based research with 67 articles (31%), followed by the *ASCE Journal of Computing in Civil Engineering* (35 articles), *Advanced Engineering Informatics* (33 articles), *Computer-aided Civil, Infrastructure Engineering* (7 articles) and *Structural Control & Health Monitoring* (7 articles).

Additionally, the minimum threshold for citation was set to 40, and thus 12 nodes out of 233 met the requirement (Fig. 6). The node size refers to the co-citation frequency of each source journal. With respect to co-citation frequency, the top five most influential journals were *Automation in Construction* (frequency = 147), *ASCE Journal of Computing in Civil Engineering* (frequency = 134), *Transactions on Pattern Analysis and Machine Intelligence* (frequency = 129), *Advanced Engineering Informatics* (frequency = 98), *Computer-aided Civil and Infrastructure Engineering* (frequency = 84). Markedly, four of these five journals were also among the top productive journals, which have made significant contributions to computer vision studies in the construction industry.

The nodes denoted by a purple ring indicate that these journals possesses a high betweenness centrality, and therefore act as a medium whereby key academics share their intellectual outputs [31]. Within a journal's co-citation network, a high betweenness centrality indicates its interdisciplinary nature [32]. In Fig. 5, the nodes highlighted by purple rings possess high levels of betweenness centrality. For example, *ASCE Journal of Construction Engineering and Management* (centrality = 0.25), *Lecture Notes in Computer Science* (centrality = 0.16), and *Computer-aided Civil and Infrastructure Engineering* (centrality = 0.15).

3.4. Keywords analysis

Keywords are representative and concise descriptions of a research article's content. They can also describe the existing research topics of a specific domain. In this paper, keywords analysis contained two parts: (1) co-occurrence; and (2) evaluation. Within the WoS database, there are two types of keywords used in the co-occurrence analysis: (1) 'author keywords'; and (2) 'keywords plus', which are identified by the journal. Both types of keywords from the 216 bibliographic records were used to generate the keyword co-occurrence network using CiteSpace, which aimed to identify the inter-closeness of research topics. Using word clouds, the keyword evaluation analysis can reflect changes within a research topic over a period of time. Word clouds are graphical representations of keyword frequency that provide greater

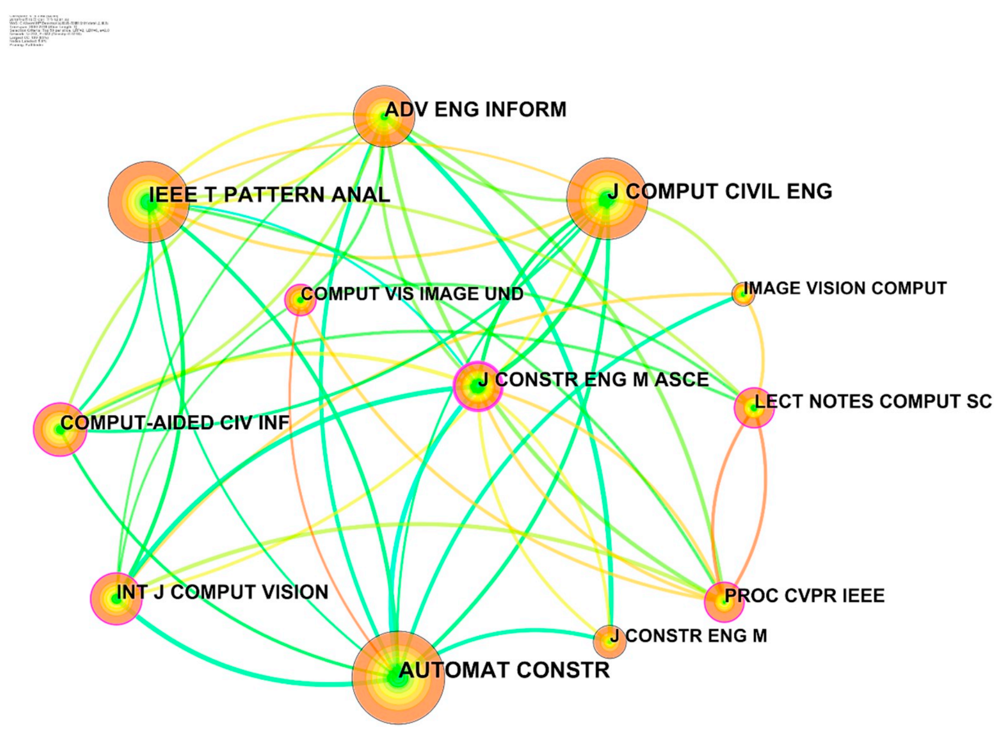


Fig. 6. Journal co-citation network.

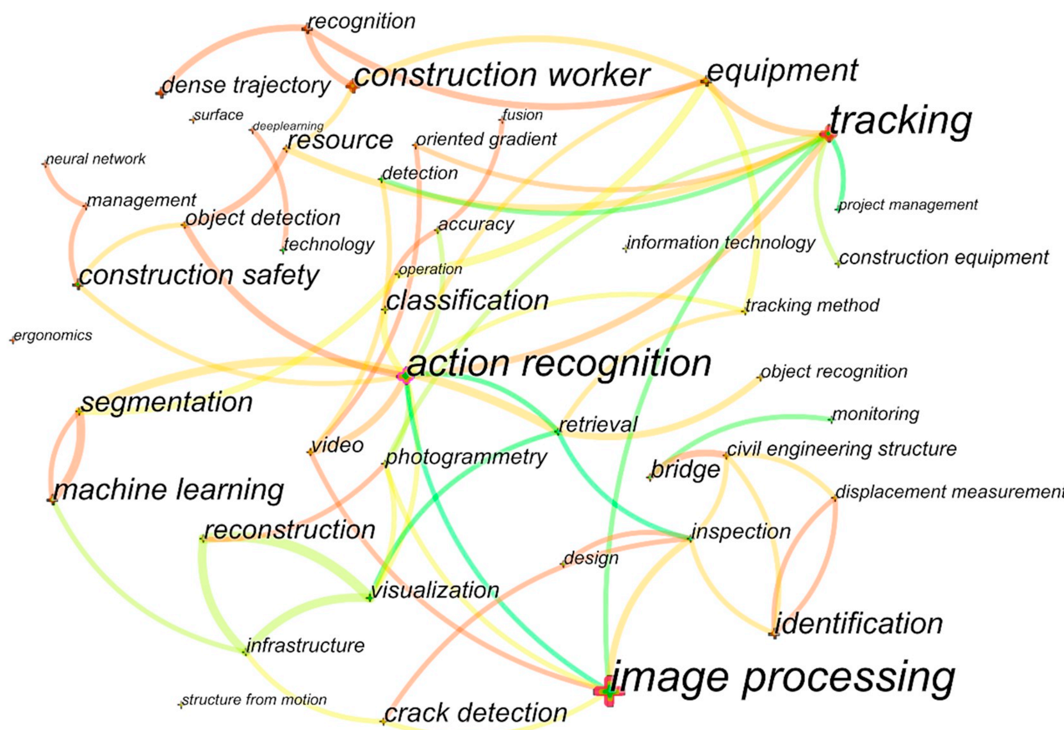


Fig. 7. Co-occurrences mapping of keywords.

prominence to words that appear more frequently [33]. In this paper, the author keywords were used to generate word clouds

3.4.1. Keyword co-occurrence

Using CiteSpace, we presented the science mapping result of the co-occurrence keywords in Fig. 7. When creating this map, we set the criterion only to include the keywords that co-occurred a minimum of three times. Some general keywords, however, were removed, such as 'computer vision', 'model', and 'construction'. The top ten keywords were displayed in Table 4. In the keyword co-occurrence network, we determined the node size by the frequency of words in the bibliometric record, and the links indicate the interrelatedness between a pair of keywords.

Determining the co-occurrence relations can provide scholars with a means to identify research topics in a specific domain. For example, in Fig. 6, the keyword of 'equipment' tended to occur with several others, such as 'tracking', 'construction worker', and 'action recognition'. The analysis revealed that equipment was mainly associated with tracking the location of equipment to improve safety or recognize its presence. In highlighting this association, we refer to the works of Memarzadeh, et al. [34] who proposed Histograms of Oriented Gradients (HOG) and Colors based algorithm to detect the presence of equipment and people on-site from video streams. The nodes with high centrality included 'image processing (centrality = 0.24)', 'action recognition (centrality = 0.22)', and 'tracking (centrality = 0.15)'. Thus, the three keywords with high centrality and frequency were: (1) 'image processing'; (2) 'action recognition'; and (3) 'tracking'. These keywords have emphasized the focus of computer vision in the field of construction.

Table 4
The top 10 frequent keywords.

Keyword	Frequency	Average published year	Centrality
Image processing	50	2010	0.24
Tracking	38	2011	0.15
Action recognition	29	2010	0.22
Construction worker	20	2016	0.02
Equipment	16	2015	0.08
Identification	13	2011	0.01
Machine learning	13	2014	0.03
Construction safety	12	2012	0.08
Segmentation	12	2014	0.05
Classification	12	2014	0.09

The co-occurrence network highlights a strong relationship between 'image processing' and 'video', 'crack detection', 'inspection', and 'tracking'. With advances in digital imaging enabled by cameras and videos, there has been an increase in their use to monitor and manage the process of construction. Furthermore, images captured from digital cameras have been used to inspect infrastructure for defects [35], tracking excavation activities [36], and performance analysis [6]. A pertinent example is the work of Yamaguchi et al. [37] who developed a crack model based on the concept of percolation. Similarly, Yu et al. [38] developed a crack detection method that was integrated with a mobile robot system to automate the inspection of concrete cracks in tunnels. In these studies, the image processing technique is a prerequisite for computer vision-based defect detection in civil infrastructures, such as template matching, histogram transforms, background subtraction, filtering, edge and boundary detection [10]. Image-processing techniques are mathematical or statistical algorithms that change the visual appearance or geometric properties of an image. The algorithms enable several pictures to be automatically processed and therefore identify and classify material clusters [39]. Color histograms, for example, can help determine similarity and dissimilarity between images. Other algorithms, such as, Gaussian or wavelet (for noise removal), Fourier analysis(low pass filtering, phase reconstruction), wavelet decomposition, Laplacian and oriented pyramids and Gabor filters can assist in obtaining explicit texture form images. Additionally, image processing is prevalent in crack detection and maintenance since it can filter the noise of images and therefore provide additional detail that is invisible to humans. In sum, image processing, therefore, can be used to recognize cracks in concrete and determine a structure's displacement.

'Tracking' referred to the use of digital cameras to track the location of resources on-sites, such as workers, equipment, and materials.

Compared with the sensor-based technology (e.g., Radio Frequency Identification, Geographical Positioning Systems, and Zigbee), the use of computer vision has several advantages over these techniques as it can provide additional information (e.g., locations, geometrical information, and behaviors) and cover large areas. Furthermore, there is no need to attach sensors and receivers on project entities [40]. Research worthy of mention here is that of Park, et al. [41] who compared various two-dimensional (2D) vision trackers to determine the most appropriate one to follow resources on congested sites. In a similar vein, Brilakis, et al. [40] developed a vision-based tracking framework that aimed to determine the spatial location of entities on a site. The tracking multiple of objects, however, remains to be a challenge due to their interacting trajectories, which can cause occlusions [27]. Moreover, determining the optimized camera location is also an issue, but when it is identified tracking accuracy can be significantly improved. For example, Zhang, et al. [42] proposed a creative approach to optimize camera locations whereby there is 100% site coverage, however, this approach is not generalizable and restricted to a specific site context.

'Action recognition' usually co-occurred with 'safety', 'video', and 'tracking'. Action recognition techniques have been applied to construction as a means to extract motion information that is necessary for automated safety monitoring. For example, a person is not in danger if the excavator is not operating even the distance between them and the excavator is small. Golparvar-Fard, et al. [3] proposed a method to recognize single actions of earthmoving equipment from site video streams based on a multiclass Support Vector Machine (SVM) classifier. The videos were represented as spatiotemporal visual features which were described with a HOG and classified using SVM.

3.4.2. Keyword evolution

Considering the limited number of published articles published between 2000 and 2010, we divided the sample of publications for analysis into three distinct periods: (1) 2000 to 2010; (2) 2011 to 2014; and (3) 2015 to 2018. The upshot was that we can see how the vernacular of computer vision research has evolved with time (Fig. 8).

The word clouds provide a quantitative and visualized method to determine the evolution of research topics (Fig. 8). What is more, the term frequency-inverse document frequency (TF-IDF) algorithm was used to identify essential keywords in the corresponding stage. It provides each keyword with a weight based on the following two criteria: (1) the frequency of its usage in the specified document (TF); and (2) the rarity of its appearance in the other documents in the corpus (IDF) [43]. The keywords with high TF-IDF for different stages are listed in Table 5. Some common keywords were removed such as 'computer vision' and 'construction'.

We can see from Table 5 that 'image processing' was a contemporary keyword. This was, however, expected as it is a fundamental component of computer vision. It also appeared from the findings displayed in Table 5 that as computer vision technologies have become more mature, there has been a subtle but distinct shift in research contents and methods.

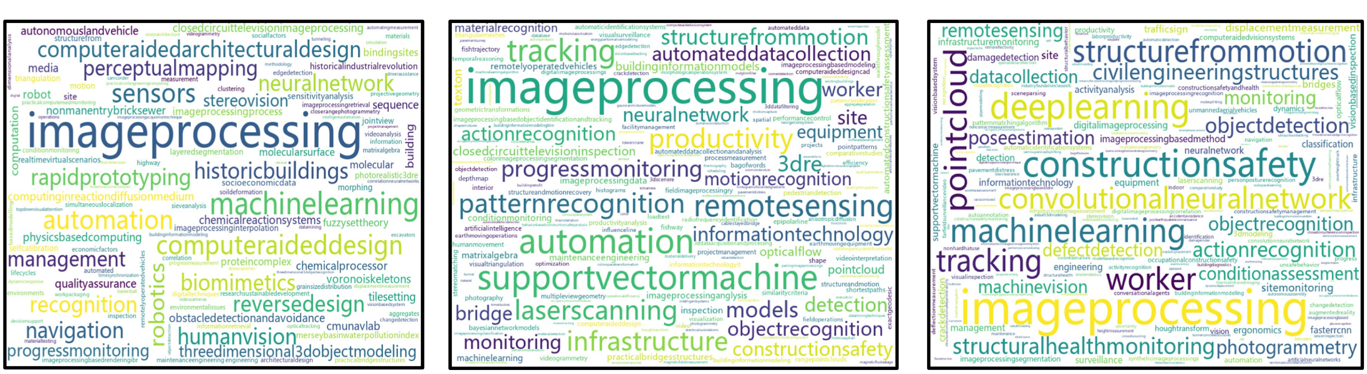
Computer vision research has paid particular attention to safety monitoring and defect detection. Seo et al. [2], for example, has been particularly influential in the area of safety monitoring as they classified computer vision-based approaches into three categories: (1) scene-based; (2) location-based; (3) action-based risk identification. This work has set the scene for a series of studies that have aimed to improve safety management on-sites [44,45]. Similarly, an influential study computer vision focusing on defect detection is Yousaf, et al. [46] where the potholes in a road were detected using SVM with an accuracy of 95.7%.

Early research computer vision methods tended to rely on the use of shallow machine learning, which used handcrafted features (Fig. 8b). Shallow machine learning methods contain two steps: (1) feature extraction and representation; and (2) classification. Some feature extractors were used to extract features from images using descriptors such as HOG [47] and Histogram of Optical Flow (HOF) [96]. These features were inputted into a shallow machine learning classifier such as SVM and k-Nearest Neighbor. For example, Park and Brilakis [47] used HOG and Haar-like features to detect individuals and equipment from images. Memarzadeh, et al. [34] developed a HOG and Color based detector to automatically detect individuals in videos. However, the use of hand-crafted features is a time-consuming and costly process.

Current research focuses on the use of deep learning methods as they have the ability to obtain robustly represent features using end-to-end learning methods, such as convolutional neural networks (CNNs) [49], Fast R-CNN [50], and Mask R-CNN [51]. The emergence of CNNs has led to the rapid developments in the area of object detection, tracking, and recognition [52]. For example, Fang, et al. [49] applied a Fast R-CNN algorithm to detect entities from images and have achieved an outstanding result with a high level of accuracy on-site (i.e., 91% and 95% for individuals and equipment, respectively).

3.5. Document analysis

We have used document co-citation and cluster analysis to objectively identify influential articles and research themes. We defined document co-citation as the frequency that two documents are cited together within others [53]. It was used to reveal the authority of references cited by selected papers. Cluster analysis is a knowledge discovery technique, which aims to determine the semantic themes hidden in the textual data [54]. A large corpus of data was classified into different clusters according to their relative degree of correlation. Based on the document co-citation network, cluster analysis was used to underpin intellectual structures of a scientific research domain, and detect research themes for future research directions [12].



(a)2000 to 2010 (b) 2011 to 2014 (c)2015 to 2018

Fig. 8. Word cloud analysis of different stages.

Table 5
Top five TF-IDF keywords.

2000 to 2010		2011 to 2014		2015 to 2018	
Keywords	TF-IDF	Keywords	TF-IDF	Keywords	TF-IDF
Image processing	0.802	Image processing	0.601	Image processing	0.290
Machine learning	0.292	Support vector machine	0.253	Construction safety	0.174
Sensors	0.219	Tracking	0.202	Deep learning	0.155
Image processing analysis	0.219	Remote sensing	0.202	Machine learning	0.135
Computer aided design	0.219	Pattern recognition	0.152	Structural health monitoring	0.100

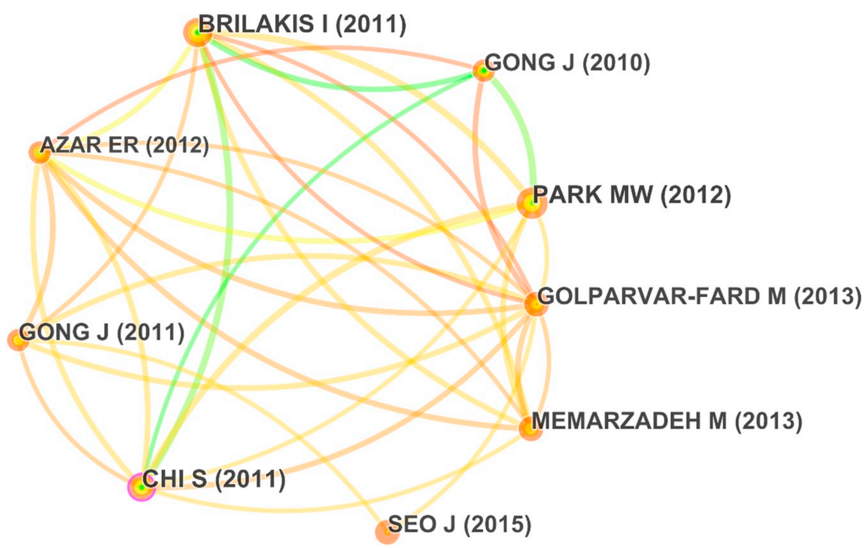


Fig. 9. Document co-citation network.

3.5.1. *Document co-citation*

In Fig. 9, the document co-citation network was presented. Setting the minimum number of document co-citations at 20 in CiteSpace, 9 out of total 288 nodes met the thresholds. Each node represents a document that is represented by the first author's name and its publication year. Each link provides the co-citation relationship between the two documents, while the size of nodes represents their co-citation frequency. We presented the top-cited documents in Table 6.

After a manual review on these high cited articles, we observed three common usages for computer vision: (1) entity detection ([47]; [55]; [34]); (2) tracking ([40]; [56]) and (3) action recognition ([3]; [57]). For example, Chi and Caldas [55] developed an approach to detect on-site objects in real-time in heavy-equipment-intensive construction sites. However, it lacks information regarding how to distinguish a worker from a pedestrian, which is a serious problem when a construction site is located in residential areas. This problem was solved by Park and Brilakis [47], who used background subtraction, the HOG, and the HSV color histogram to detect construction workers by extracting three features from images: motion, shape, and color.

Some computer-vision supported applications can be applied in practice, such as safety monitoring [2] and used for productivity analysis [6]. For example, Gong and Caldas [56] developed a prototype system to automated measure productivity from videos. Seo, et al. [2] presented a systematic review on computer vision-based safety monitoring, in which detailed object detection, tracking, and recognition techniques were discussed. Additionally, the works of Chi and Caldas [55] and Gong and Caldas [6] have a high centrality 0.10, which suggest they have made a source of reference for a vast number of studies.

3.5.2. *Cluster analysis*

In Fig. 10, a total of 13 clusters were identified through the log-likelihood ratio (LLR) algorithm, which was embedded in CiteSpace [31]. A label for each identified cluster was generated using the LLR algorithm to reveal the main research content and generate a label for the corresponding cluster using keywords derived from the cited documents. The quality of cluster labeling depends on the variety,

Table 6
High cited articles.

Title	Frequency	Centrality	Reference
Construction worker detection in video frames for initializing vision trackers	27	0.03	[47]
Automated vision tracking of project related entities	26	0.04	[40]
Automated Object Identification Using Optical Video Cameras on Construction Sites	24	0.10	[55]
Automated 2D detection of construction equipment and workers from site video streams using histograms of oriented gradients and colors	23	0.03	[34]
Computer Vision-Based Video Interpretation Model for Automated Productivity Analysis of Construction Operations	22	0.10	[6]
An object recognition, tracking, and contextual reasoning-based video interpretation method for rapid productivity analysis of construction operations	22	0.06	[56]
Vision-based action recognition of earthmoving equipment using spatio-temporal features and support vector machine classifiers	21	0.02	[3]
Computer vision techniques for construction safety and health monitoring	21	0.02	[2]
Automated Visual Recognition of Dump Trucks in Construction Videos	19	0.06	[57]

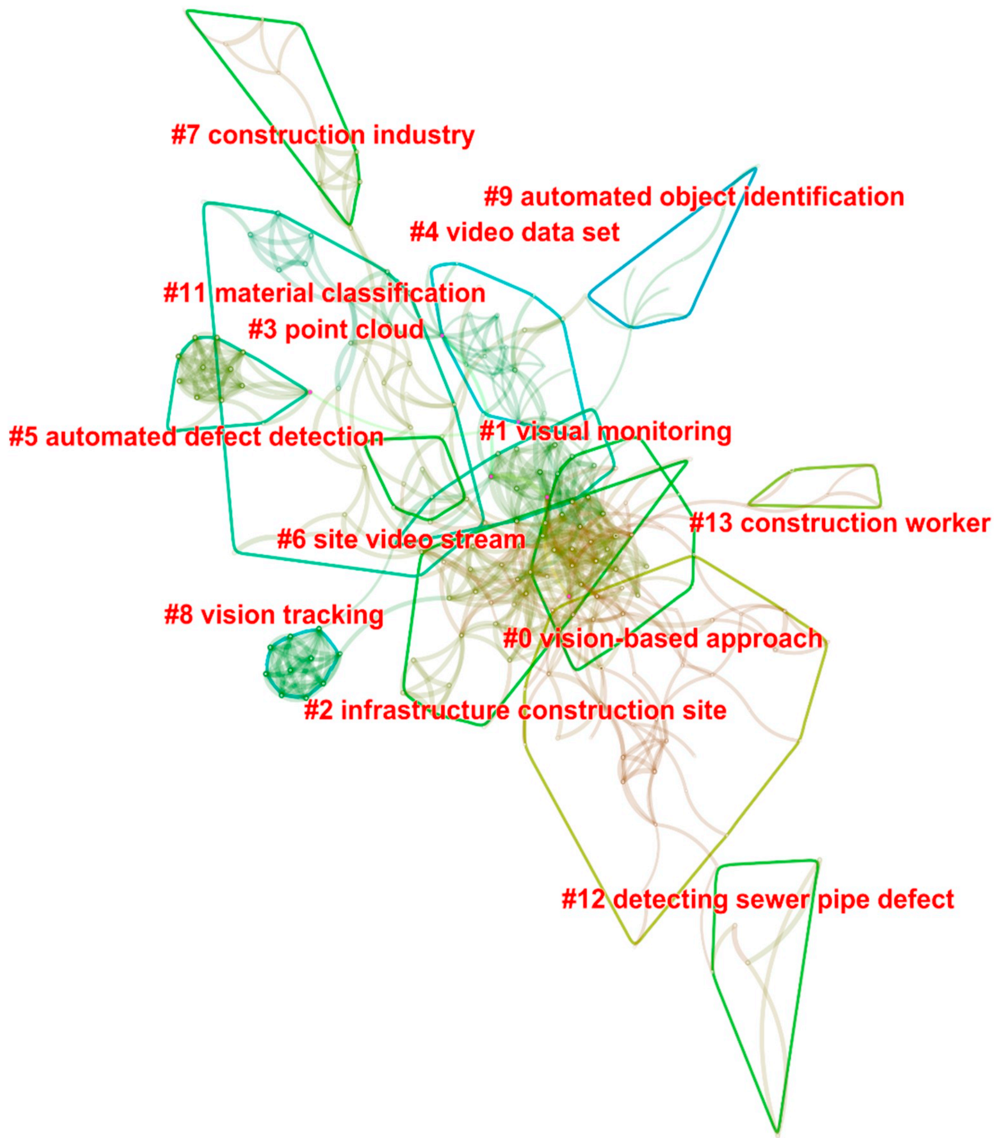


Fig. 10. The clustering results in LLR algorithm.

breadth, and depth of the set of terms derived from keywords within articles [58]. Notably, a critical review was needed to interpret the clustering labels and summarize research themes based on articles contained within each cluster. For instance, cluster 5 ('Automated defect detection') and cluster 12 ('Detecting sewer pipe defect') revealed the same research theme.

In Table 7, we presented detailed information about the clusters, which included silhouette, cluster labels, alternative labels, and size (i.e., the number of members). The silhouette value, ranging from −1 to 1, indicates the uncertainty that we need to take into account when interpreting the nature of the cluster [58]. It reflects the average homogeneity of a cluster [59]. The value of 1 means a perfect

Table 7
Documents co-citation clusters.

Cluster ID	Size	Silhouette	Cluster label (LLR)	Alternative label	Mean Year	Reference
#0	38	0.778	Vision-based approach	Convolutional neural network; Safety	2015	[45]
#1	32	0.727	Visual monitoring	Equipment; Equipment actins	2011	[60]
#2	27	0.744	Infrastructure construction site	Construction equipment; Point cloud	2011	[61]
#3	26	0.776	Point cloud	Equipment; Dataset	2008	[62]
#4	21	0.639	Video data set	Construction worker; Productivity	2006	[63]
#5	13	0.994	Automated defect detection	Support vector machine; Sewer inspection	2008	[64]
#6	13	0.861	Site video stream	Equipment; Path conflict	2008	[40]
#7	10	0.98	Construction industry	Point clouds; 3D reconstruction	2011	[65]
#8	10	0.992	Vision tracking	Tracking tag; Facility operations	2006	[41]
#9	8	0.907	Automated object identification	Construction site; Technologies	2005	[66]
#11	7	0.904	Material classification	Construction material; Building element	2011	[67]
#12	6	0.972	Detecting sewer pipe defect	Detection accuracy; Data size	2012	[10]
#13	5	0.985	Construction worker	Objects; Hardhats	2014	[68]

separation from other clusters. The clustering result has high reliability when the silhouette value of a cluster exceeds 0.7 [58]. In CiteSpace, the representative papers were identified based on the metric of 'coverage' which was calculated using Eq. (2):

$$\text{Coverage} = \frac{N}{M}, (0 < Coverage \leq 1) \tag{2}$$

where N is the number of cited articles in the corresponding cluster, and M is the number of all articles in the corresponding cluster. For example, in Fig. X, M of cluster #0 is 38, and the N of Fang, et al. [45] is 14. Thus, the coverage of Fang, et al. [45] is 0.37. Based on the 'coverage', the most representative article of each cluster was listed in Table 7.

4. Qualitative interpretation

Our qualitative interpretation of the literature provided a contextual backdrop to the science mapping and further examined the research themes, identified knowledge gaps, and proposed a framework for future research.

4.1. Summary of research themes

Computer vision research published up until 2018 has focused the object recognition and tracking people and plant on-site (e.g., cluster #2 and #6) and the status and quality of infrastructure (e.g., cluster #5 and 12). Based on the results of the cluster analysis, we examined these themes in further detail.

4.1.1. Computer vision on-site

On-site the focus of computer vision has been on monitoring safety with regard to people's behavior and site condition (Table 8) and in the context of performance tracking on project and operational monitoring levels (Table 9):

1. *Safety monitoring*: Han and Lee [69], for example, developed a computer vision-based framework to monitor unsafety behaviors, which contained four parts: (1) The identification of unsafety behaviors from safety documents and historical records; (2) the use of a laboratory experiment to collect motion templates for the identified unsafe actions; (3) the extraction of 3D skeleton from videos; and (4) real-time detection of unsafe behaviors using video.
2. *Performance monitoring*: At a project-level emphasis has been placed on tracking the progress of construction using a series of metrics such as a Schedule Performance Index (SPI). The operational level of attention has concentrated on the productivity analysis of individuals or equipment where non-valuing adding activities are measured (e.g., waiting, idle, excessive travel, and transporting time).

Golparvar-Fard et al. [73] examined the status of construction by comparing its 'as-planned' with the 'as-actual' states using 2D time-lapse photographs. In this instance, the time-lapse images were used to document the work-in-progress, which was compared to a four-dimensional building information model (BIM). Golparvar-Fard et al. [74] developed a pipeline of Multi-View Stereo and voxel coloring algorithms to improve the density of 3D point clouds and presented a method for superimposing them within a BIM.

Computer vision has also been used to determine productivity [56]. The extent of resources that were being utilized is examined against crew-balance charts. For example, an automated interpretation technique can be used to extract and measure productivity information from the video. The interpretation progress contains computer vision, reasoning, and multimedia methods. More specifically, computer vision was used to recognize and track objects autonomously in a video. The number of objects to be tracked can be minimized by selecting an

Table 8
Prior works on computer vision-based safety monitoring.

Research theme	Types of hazards	Problem	Types of data resource	Descriptions	Reference
Safety Monitoring	Unsafe Behaviors	Failure to use PPE	Not wearing hardhat in working areas	2D Images	Safety Monitoring	[44]
			Motion recognition for tracking unsafe actions	Video	HOG descriptor was used to extract skeleton models and kernel Principal Component Analysis was used to analyze 3D skeletons	[69]
		Abnormal operation of individuals	Cross structural support (e.g., concrete and steel)	2D image	A Mask Region Based CNN was used to recognize the relationship between people and concrete/steel supports	[51]
			Falling when climbing and dismounting from a ladder	Video	A hybrid deep learning model was developed to identify unsafe actions by integrating CNN and Long Short Term Memory (LSTM)	[70]
		Entering hazardous area	Struck or be close to bulldozer/excavator that is backing up	Video	Applied computer vision technique to extract spatial information for further fuzzy inference to assess safety levels of each entity	[71]
	Unsafe Conditions	Spatial conflicts	Blind lifts on congested offshore platform environments	Laser scanner, sensors	Developed a framework for real-time pro-active safety assistance for mobile crane lifting operation	[72]

Table 9
Prior works on computer vision-based performance monitoring.

Research theme		Objects	Problem	Types of data resource	Technique	Descriptions	Reference
Performance monitoring	Tracking project-level information	Tracking progress on construction sites	Occupancy-based progress assessment	2D images; Time-lapse images	Point clouds; imaging processing;	Superimposing the as-built model in point clouds with as-planed BIM, and reasoning the occupancy	[74]
			Appearance-based progress assessment	2D images; Time-lapse images		Observing the appearance of the BIM elements to reason the progress	[76]
	Monitoring operation-level information	Operation analysis of construction workers	Productivity measurement; Idle time detection	Video streams	Human pose analyzing algorithms	Automated productivity measurement through human pose analyzing algorithms	[75]
		Operation analysis of construction equipment	Productivity measurement; Cyclic operation analysis; Idle time detection	Video streams	Object detection and tracking, and activity recognition algorithm (e.g., SVM)	Analyze equipment in the earthworks of excavation, material hauling, and dirt loading	[56]

algorithm that was linked to a domain knowledge (e.g., Method Productivity Delay Model) [56]. For example, Peddi et al. [75] measured the productivity from videos by analyzing the pose of people while they were tying rebar with the results of 85% to 89% accuracy.

In addition to safety monitoring and performance analysis, there have been several studies that have focused on the automatic generation of BIMs [63], quality inspection of steel bars [77], and detecting materials to be recycled [78].

4.1.2. Computer vision and infrastructure

Traditionally, the inspection and assessment of infrastructure have been manually performed by qualified structural engineers who seek to identify defects (e.g., cracking, delamination, and spalling) and then measure their effect (e.g., depth, width, and length) [10]. It has been widely recognized that computer vision juxtaposed with other technologies such as unmanned aerial vehicles (UAV), 3D digital image correlation technique, and Closed Circuit Television (CCTV) imaging, can perform these tasks as well as inspect the structural integrity of tunnels [79], crack detection [80] and the structural condition of bridges [81]. A summary of key computer vision research that has examined infrastructure is presented in Table 10.

4.2. Knowledge gaps

4.2.1. Lack of an adequately sized database

Machine learning algorithms are dependent on the quantity and quality of information used to train them. Whether the identification of hazards on a construction site or the detection of structural defects, an extensive and high-quality database of images is a pre-requisite to ensure the successful application of computer vision. As evident from the studies undertaken to date, there is an absence of adequately sized databases that can be used to ensure the accuracy of computer vision. Thus, the limitation of adequately sized datasets is inhibiting the development of computer vision in construction. Moreover, there appears to be a reluctance of researchers to share their training sets. It is therefore vital that journal editors require all papers that are accepted for publication to provide a cop the training sets used and even datasets used for a particular study. Though, privacy laws can prevent this from occurring.

In the meantime, researchers reliant on using small databases will need to use data augmentation techniques. In this instance, where minor alterations to the existing data were undertaken such as image rotation, flipping, and random cutting [85]. Nevertheless, this progress may lead to potential loss of relevant data or outliers needed for training. With limited data, researchers tend to choose a relatively small sample to undertake their experimental works, which renders it difficult to compare and contrast evaluation metrics such as precision and recall.

4.2.2. Data privacy

We acknowledge that freely accessible databases are costly and timely to construct and may also contain private and sensitive information. Furthermore, they may be challenging to apply in different countries, and prevailing privacy laws may prevent the sharing of data. For instance, Europe has enacted regulations on data privacy protection referred to as the General Data Protection Regulation (GDPR) [86]. This regulation provides citizens of the European Union with rights when companies or institutions process personal data.

While computer vision has enabled headway to be made in identifying individuals who have performed an unsafe act on-site using videos cameras, it can be viewed as violating a person's privacy if they have not agreed to be monitored. Data acquisition equipment cannot be installed on a construction site if the people do not agree [87]. Monitoring devices can make people uncomfortable and even generate negative emotions. Furthermore, it may restrict creative behaviors if people realize their actions are monitored [88].

Table 10
Prior works on computer vision-based structural detection.

Research theme	Problem	Types of data resource	Techniques	Descriptions	Reference
Defect detection and condition assessment on infrastructures (Bridges, tunnels, underground pipe, and asphalt pavements)	Cracking detection	2D Images	Image processing algorithm (e.g., Histogram-based classification algorithm and support vector machines (SVM)); Image capture technique (e.g., aerial robots)	Applied algorithm (e.g., SVM) to detect cracks on a concrete deck surface	[82]
	Delamination/ spalling/holes	2D images	Pattern recognition approach (Segmentation, template matching, and morphological pre-processing)	Combined segmentation, template matching and morphological pre-processing for spall detection and assessment on concrete columns	[83]
	Other structural defects	2D images	Computer vision techniques and sensing technology	Integrated video imagery and bridge responses to detect loss of connectivity between different composite sections.	[84]

4.2.3. Technical challenges

Computer vision-based research comprises of two core steps: (1) data collection (e.g. 2D images, time-lapse images, and videos); and (2) analysis. In the case of data collection, the positioning and orientation of cameras need to be given consideration in order to capture the appropriate images of objects. Computer vision methods obey the principle 'what you see is what you can analyze' [49]. Thus, the quality of data collected is critical so that it can be effectively analyzed and used to accurately detect an object. Several factors can hinder the accuracy of object detection on-site, including poor lighting, cluttered backgrounds, and occlusions. As a result, there is a need for a multitude of camera positions to be placed on a site to overcome such problems.

The analysis of data analysis can be undertaken using several approaches but the most common are either conventional shallow learning methods such as SVM [3,34] or deep learning that utilizes CNNs [49]. As we mentioned above deep learning, particularly CNN's and Recurrent Neural Networks are becoming an increasingly popular method for image classification and object detection due to their ability to automatically extract features [44,89]. While deep learning is being widely used, there are several technical challenges that confront its use in practice. First, deep learning can only learn from the correlation between input and output and is not able to determine causality. In the case of safety monitoring, for example, not only is there a need to identify individuals and working conditions, but also the interactions between them. To date, this interaction (as identified in Section 3.4.1) has not to be examined and thus needs to be a future line of inquiry. Second, there is an absence of a generic model that can be used to address a multitude of problems. Models have been developed and trained to tackle a specific problem scenario. In practice, if such models are to be effective, they will need to identify a wide range of tasks, which will require us to develop new algorithms to fulfil this requirement.

4.2.4. Semantic gap

There is a 'semantic gap' between the low-level feature extracted from the images by computer vision algorithms and the high-level semantic meaning that people recognize from an image [90]. As a result of this semantic gap, developments in automated computer vision may be stymied. In the case of hazards identification, for example, not only do objects need to be detected from the images, but also domain knowledge, This domain knowledge is needed to provide a context within the safety regulations [50].

Further research, therefore, could integrate ontology and computer vision techniques to address the semantic gap. Ontology is a popular approach applied for modeling information due to its computer-oriented and logic-based features, which provides a way to formally represent domain knowledge by the explicit definition of classes, relationships, functions, axioms, and instances [91,92]. It can represent knowledge with explicit and rich semantics, which can enable knowledge query and reasoning to be performed [93].

A framework combining ontology and computer vision techniques could be developed specifically focusing on developing:

- domain knowledge that is formally represented by an ontology model where different rules can be encoded for the specific application, such as hazard reasoning and defect identification;
- spatial relationships between objects, which can be automated detected using computer vision algorithms [51]; and
- in conjunction with the ontology a specific rule engine such as Drools [94]. In this instance, the domain knowledge can be used to accommodate the detected objects and their relationships (e.g., automated hazard reasoning or various structural defect recognition).

With the addition of a domain knowledge represented by ontologies, the ability of computer vision to understand scenes from images can be further improved.

5. Conclusion

Computer vision has attracted an increasing amount of attention from researchers and practitioners. We have undertaken a detailed review of the extant literature using a science mapping approach to: (1) determine the collaboration of authors and institutions, and indicate the influential journals;(2) identify the keywords used within the domain of computer vision; (3) identify high cited articles and create different clusters with labels to objectively reflect the emerging research themes; and (4) categorize the main research themes so that gaps in knowledge can be identified.

We identified three research communities and leading figureheads in the field of computer vision in construction: (1) Brilakis Ioannis K., (2) Li Heng, and (3) Ding Lieyun. The most productive authors were Brilakis Ioannis K, Li Heng, and Park Manwoo Mani. Our analysis revealed that most computer vision based articles emanated from the Georgia Institute of Technology and Hong Kong Polytechnic University.

While considerable headway has been made to apply computer vision to construction in areas such as safety monitoring, productivity analysis, and determining structural defects, several technical challenges hindered its development. We have observed that there is a paucity of adequately sized databases for training and issues associated with data privacy can prevent them from being shared. Juxtaposed with the technical issues that we have identified (e.g., an inability to identify causality and the need for new algorithms to identify simultaneously identify several tasks) the field of computer vision in construction remains a fertile line of inquiry.

Despite the contribution of this paper, the findings still have limitations. First, in this paper, two metrics provided by CiteSpace were used to find the key nodes in each network: betweenness centrality and burst strength. More metrics should be considered in future studies, such as the negative degree, eigenvector centrality, clustering coefficient, and average neighbor degree [95]. Second, the literature sample was limited to English journal articles, which might exclude studies that have been published in other languages.

Acknowledgments

This research is partly supported by “National Natural Science Foundation of China” (No. 51878311, No. 71732001, No. 71821001).

References

[1] M. Sonka, V. Hlavac, R. Boyle, Image Processing, Analysis and Machine Vision, 3 rd. ed., Springer, Boston, 2008, https://doi.org/10.1007/978-1-4899-3216-7.
[2] J. Seo, S. Han, S. Lee, H. Kim, Computer vision techniques for construction safety and health monitoring, Adv. Eng. Inform. 29 (2) (2015) 239–251, https://doi.org/10.1016/j.aei.2015.02.001.
[3] M. Golparvar-Fard, A. Heydarian, J.C. Niebles, Vision-based action recognition of earthmoving equipment using spatio-temporal features and support vector machine classifiers, Adv. Eng. Inform. 27 (4) (2013) 652–663, https://doi.org/10.1016/j.aei.2013.09.001.
[4] M.W. Park, I. Brilakis, Continuous localization of construction workers via integration of detection and tracking, Autom. Constr. 72 (2016) 129–142, https://doi.org/10.1016/j.autcon.2016.08.039.
[5] K.K. Han, M. Golparvar-Fard, Appearance-based material classification for monitoring of operation-level construction progress using 4D BIM and site photologs, Autom. Constr. 53 (2015) 44–57, https://doi.org/10.1016/j.autcon.2015.02.007.
[6] J. Gong, C.H. Caldas, Computer vision-based video interpretation model for automated productivity analysis of construction operations, J. Comput. Civ. Eng. 24 (3) (2010) 252–263, https://doi.org/10.1061/(asce)cp.1943-5487.0000027.
[7] Q. Fang, H. Li, X.C. Luo, L.Y. Ding, H.B. Luo, C.Q. Li, Computer vision aided inspection on falling prevention measures for steeplejacks in an aerial environment, Autom. Constr. 93 (2018) 148–164, https://doi.org/10.1016/j.autcon.2018.05.022.
[8] B.E. Mneymneh, M. Abbas, H. Khoury, Evaluation of computer vision techniques for automated hardhat detection in indoor construction safety applications, Frontiers of Engineering Management 5 (2) (2018) 227–239, https://doi.org/10.15302/j-fem-2018071.
[9] J. Seo, K. Yin, S. Lee, Automated postural ergonomic assessment using a computer vision-based posture classification, Construction Research Congress (2016), https://doi.org/10.1061/9780784479827.082.
[10] C. Koch, K. Georgieva, V. Kasireddy, B. Akinci, P. Fieguth, A review on computer vision based defect detection and condition assessment of concrete and asphalt civil infrastructure, Adv. Eng. Inform. 29 (2) (2015) 196–210, https://doi.org/10.1016/j.aei.2015.01.008.
[11] M.R. Hosseini, I. Martek, E.K. Zavadskas, A.A. Aibinu, M. Arashpour, N. Chileshe, Critical evaluation of off-site construction research: a scientometric analysis, Autom. Constr. 87 (2018) 235–247, https://doi.org/10.1016/j.autcon.2017.12.002.
[12] J. Song, H. Zhang, W. Dong, A review of emerging trends in global PPP research: analysis and visualization, Scientometrics 107 (3) (2016) 1111–1147, https://doi.org/10.1007/s11192-016-1918-1.
[13] X. Zhao, A scientometric review of global BIM research: analysis and visualization, Autom. Constr. 80 (2017) 37–47, https://doi.org/10.1016/j.autcon.2017.04.002.
[14] R. Jin, S. Gao, A. Cheshmehzangi, E. Aboagye-Nimo, A holistic review of off-site construction literature published between 2008 and 2018, J. Clean. Prod. 202 (2018) 1202–1219, https://doi.org/10.1016/j.jclepro.2018.08.195.
[15] T.O. Olawumi, D.W.M. Chan, J.K.W. Wong, Evolution in the intellectual structure of BIM research: a bibliometric analysis, J. Civ. Eng. Manag. 23 (8) (2017) 1060–1081, https://doi.org/10.3846/13923730.2017.1374301.
[16] A.R. Ramos-Rodriguez, J. Ruiz-Navarro, Changes in the intellectual structure of strategic management research: a bibliometric study of the Strategic Management Journal, 1980–2000, Strateg. Manag. J. 25 (10) (2004) 981–1004, https://doi.org/10.1002/smj.397.
[17] L.B. De Rezende, P. Blackwell, M.D. Pessanha Gonçalves, Research focuses, trends, and major findings on project complexity: a bibliometric network analysis of 50 years of project complexity research, Proj. Manag. J. 49 (1) (2018) 42–56, https://doi.org/10.1177/875697281804900104.
[18] D. Neufeld, Y. Fang, S.L. Huff, The IS identity crisis, Commun. Assoc. Inf. Syst. 19 (2007), https://doi.org/10.17705/1cais.01919.
[19] A. Serenko, N. Bontis, L. Booker, K. Sadeddin, T. Hardie, A scientometric analysis of knowledge management and intellectual capital academic literature (1994-2008), J. Knowl. Manag. 14 (1) (2010) 3–23, https://doi.org/10.1108/13673271011015534.
[20] C. Chen, CiteSpace: a practical guide for mapping scientific literature, Hauppauge, N.Y, Nova Science 169 (2016), https://doi.org/10.22201/iibi.24488321xe.2017.nesp1.57894.
[21] C. Chen, Science mapping: a systematic review of the literature, Journal of Data and Information Science 2 (2) (2017) 1–40, https://doi.org/10.1515/jdis-2017-0006.
[22] C. Chen, Z. Hu, S. Liu, H. Tseng, Emerging trends in regenerative medicine: a scientometric analysis in CiteSpace, Expert. Opin. Biol. Ther. 12 (5) (2012) 593–608, https://doi.org/10.1517/14712598.2012.674507.
[23] L.C. Freeman, A set of measures of centrality based on betweenness, Sociometry 40 (1) (1977) 35–41, https://doi.org/10.2307/3033543.
[24] J. Kleinberg, K. Discovery, Bursty and hierarchical structure in streams, Data Min. Knowl. Disc. 7 (4) (2003) 373–397, https://doi.org/10.1145/775060.775061.
[25] H. Son, C. Kim, 3D structural component recognition and modeling method using color and 3D data for construction progress monitoring, Autom. Constr. 19 (7) (2010) 844–854, https://doi.org/10.1016/j.autcon.2010.03.003.
[26] Y.H. Wu, H. Kim, C. Kim, S.H. Han, Object recognition in construction-site images using 3D CAD-based filtering, J. Comput. Civ. Eng. 24 (1) (2010) 56–64, https://doi.org/10.1061/(asce)0887-3801(2010)24:1(56).
[27] J. Yang, O. Arif, P.A. Vela, J. Teizer, Z.K. Shi, Tracking multiple workers on construction sites using video cameras, Adv. Eng. Inform. 24 (4) (2010) 428–434, https://doi.org/10.1016/j.aei.2010.06.008.
[28] J.S. Katz, B.R. Martin, What is research collaboration? Res. Policy 26 (1) (1997) 1–18, https://doi.org/10.1016/S0048-7333(96)00917-1.
[29] A.E. Bayer, J.C. Smart, G.W. McLaughlin, Mapping intellectual structure of a scientific subfield through author cocitations, Journal of the American Society for Information Science banner 41 (6) (1990) 444, https://doi.org/10.1002/(sici)1097-4571(199009)41:6<444::aid-asi12>3.0.co;2-j.
[30] M. Muja, D. Lowe, Fast approximate nearest neighbors with automatic algorithm configuration, VISAPP International Conference on Computer Vision Theory and Applications 2, 2 2009, pp. 331–340, , https://doi.org/10.5220/0001787803310340.
[31] T.O. Olawumi, D. Chan, A scientometric review of global research on sustainability and sustainable development, J. Clean. Prod. 183 (2018) 231–250, https://doi.org/10.1016/j.jclepro.2018.02.162.
[32] L. Leydesdorff, Betweenness centrality as an indicator of the interdisciplinarity of scientific journals, J. Am. Soc. Inf. Sci. Technol. 58 (9) (2007) 1303–1319, https://doi.org/10.1002/asi.20614.
[33] R. Riggs, S. Hu, Disassembly liaison graphs inspired by word clouds, Procedia CIRP 7 (2013) 521–526, https://doi.org/10.1016/j.procir.2013.06.026.
[34] M. Memarzadeh, M. Golparvar-Fard, J. Niebles, Automated 2D detection of construction equipment and workers from site video streams using histograms of oriented gradients and colors, Autom. Constr. 32 (2013) 24–37, https://doi.org/10.1016/j.autcon.2012.12.002.
[35] N. Metni, T. Hamel, A UAV for bridge inspection: visual servoing control law with orientation limits, Autom. Constr. 17 (1) (2007) 3–10, https://doi.org/10.1016/j.autcon.2006.12.010.
[36] C.A. Quinones-Rozo, Y.M.A. Hashash, L.Y. Liu, Digital image reasoning for tracking excavation activities, Autom. Constr. 17 (5) (2008) 608–622, https://doi.org/10.1016/j.autcon.2007.10.008.
[37] T. Yamaguchi, S. Hashimoto, Fast crack detection method for large-size concrete surface images using percolation-based image processing, Mach. Vis. Appl. 21 (5)

(2010) 797–809, https://doi.org/10.1007/s00138-009-0189-8.
[38] S.-N. Yu, J.-H. Jang, C.-S. Han, Auto inspection system using a mobile robot for detecting concrete cracks in a tunnel, Autom. Constr. 16 (3) (2007) 255–261, https://doi.org/10.1016/j.autcon.2006.05.003.
[39] I.K. Brilakis, L. Soibelman, Y. Shinagawa, Construction site image retrieval based on material cluster recognition, Adv. Eng. Inform. 20 (4) (2006) 443–452, https://doi.org/10.1016/j.aei.2006.03.001.
[40] I. Brilakis, M.W. Park, G. Jog, Automated vision tracking of project related entities, Adv. Eng. Inform. 25 (4) (2011) 713–724, https://doi.org/10.1016/j.aei.2011.01.003.
[41] M.W. Park, A. Makhmalbaf, I. Brilakis, Comparative study of vision tracking methods for tracking of construction site resources, Autom. Constr. 20 (7) (2011) 905–915, https://doi.org/10.1016/j.autcon.2011.03.007.
[42] Y. Zhang, H. Luo, M. Skitmore, Q. Li, B. Zhong, Optimal camera placement for monitoring safety in metro station construction work, J. Constr. Eng. Manag. 145 (1) (2018) 04018118, , https://doi.org/10.1061/(asce)co.1943-7862.0001584.
[43] A. Khan, B. Baharudin, L.H. Lee, K. Khan, A review of machine learning algorithms for text-documents classification, Journal of Advances in Information Technology 1 (1) (2010) 4–20, https://doi.org/10.4304/jait.1.1.4-20.
[44] Q. Fang, H. Li, X.C. Luo, L.Y. Ding, H.B. Luo, T.M. Rose, W.P. An, Detecting non-hardhat-use by a deep learning method from far-field surveillance videos, Autom. Constr. 85 (2018) 1–9, https://doi.org/10.1016/j.autcon.2017.09.018.
[45] W.L. Fang, L.Y. Ding, H.B. Luo, P.E.D. Love, Falls from heights: a computer vision-based approach for safety harness detection, Autom. Constr. 91 (2018) 53–61, https://doi.org/10.1016/j.autcon.2018.02.018.
[46] M.H. Yousaf, K. Azhar, F. Murtaza, F. Hussain, Visual analysis of asphalt pavement for detection and localization of potholes, Adv. Eng. Inform. 38 (2018) 527–537, https://doi.org/10.1016/j.aei.2018.09.002.
[47] M.-W. Park, I. Brilakis, Construction worker detection in video frames for initializing vision trackers, Autom. Constr. 28 (2012) 15–25, https://doi.org/10.1016/j.autcon.2012.06.001.
[49] W. Fang, L. Ding, B. Zhong, P.E. Love, H. Luo, Automated detection of workers and heavy equipment on construction sites: a convolutional neural network approach, Adv. Eng. Inform. 37 (2018) 139–149, https://doi.org/10.1016/j.aei.2018.05.003.
[50] Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, T.M. Rose, W. An, Detecting non-hardhat-use by a deep learning method from far-field surveillance videos, Autom. Constr. 85 (2018) 1–9, https://doi.org/10.1016/j.autcon.2017.09.018.
[51] W. Fang, B. Zhong, N. Zhao, P.E. Love, H. Luo, J. Xue, S. Xu, A deep learning-based approach for mitigating falls from height with computer vision: convolutional neural network, Adv. Eng. Inform. 39 (2019) 170–177, https://doi.org/10.1016/j.aei.2018.12.005.
[52] A. Krizhevsky, I. Sutskever, G.E. Hinton, Imagenet classification with deep convolutional neural networks, Commun. ACM 60 (6) (2017) 84–90, https://doi.org/10.1145/3065386.
[53] H. Small, Co-citation in the scientific literature: a new measure of the relationship between two documents, J. Am. Soc. Inf. Sci. 24 (4) (1973) 265–269, https://doi.org/10.1002/asi.4630240406.
[54] M. Hossain, V.R. Prybutok, N. Evangelopoulos, Causal latent semantic analysis (cLSA): An illustration, Int. Bus. Res. (2011), https://doi.org/10.5539/ibr.v4n2p38.
[55] S. Chi, C.H. Caldas, Automated object identification using optical video cameras on construction sites, Computer-Aided Civil and Infrastructure Engineering 26 (5) (2011) 368–380, https://doi.org/10.1111/j.1467-8667.2010.00690.x.
[56] J. Gong, C.H. Caldas, An object recognition, tracking, and contextual reasoning-based video interpretation method for rapid productivity analysis of construction operations, Autom. Constr. 20 (8) (2011) 1211–1226, https://doi.org/10.1016/j.autcon.2011.05.005.
[57] E.R. Azar, B. McCabe, Automated visual recognition of dump trucks in construction videos, J. Comput. Civ. Eng. 26 (6) (2012) 769–781, https://doi.org/10.1061/(asce)cp.1943-5487.0000179.
[58] C. Chen, F. Ibekwe-SanJuan, J. Hou, The structure and dynamics of cocitation clusters: a multiple-perspective cocitation analysis, Journal of the American Society for Information Science and Technology Banner 61 (7) (2010) 1386–1409, https://doi.org/10.1002/asi.21309.
[59] P.J. Rousseeuw, Silhouettes: a graphical aid to the interpretation and validation of cluster analysis, J. Comput. Appl. Math. 20 (1987) 53–65, https://doi.org/10.1016/0377-0427(87)90125-7.
[60] J. Yang, M.-W. Park, P.A. Vela, M. Golparvar-Fard, Construction performance monitoring via still images, time-lapse photos, and video streams: now, tomorrow, and the future, Adv. Eng. Inform. 29 (2) (2015) 211–224, https://doi.org/10.1016/j.aei.2015.01.011.
[61] J. Teizer, Status quo and open challenges in vision-based sensing and tracking of temporary resources on infrastructure construction sites, Adv. Eng. Inform. 29 (2) (2015) 225–238, https://doi.org/10.1016/j.aei.2015.03.006.
[62] I. Brilakis, H. Fathi, A. Rashidi, Progressive 3D reconstruction of infrastructure with videogrammetry, Autom. Constr. 20 (7) (2011) 884–895, https://doi.org/10.1016/j.autcon.2011.03.005.
[63] I. Brilakis, M. Lourakis, R. Sacks, S. Savarese, S. Christodoulou, J. Teizer, A. Makhmalbaf, Toward automated generation of parametric BIMs based on hybrid video and laser scanning data, Adv. Eng. Inform. 24 (4) (2010) 456–465, https://doi.org/10.1016/j.aei.2010.06.006.
[64] M.R. Halfawy, J. Hengmeechai, Automated defect detection in sewer closed circuit television images using histograms of oriented gradients and support vector machine, J. Comput. Civ. Eng. 38 (2014) 1–13, https://doi.org/10.1061/(asce)cp.1943-5487.0000312.
[65] H. Fathi, F. Dai, M. Lourakis, Automated as-built 3D reconstruction of civil infrastructure using computer vision: achievements, opportunities, and challenges, Adv. Eng. Inform. 29 (2) (2015) 149–161, https://doi.org/10.1016/j.aei.2015.01.012.
[66] S. Chi, C. Caldas, Automated object identification using optical video cameras on construction sites, Computer-Aided Civil and Infrastructure Engineering 26 (5) (2011) 368–380, https://doi.org/10.1111/j.1467-8667.2010.00690.x.
[67] A. Dimitrov, M. Golparvar-Fard, Vision-based material recognition for automated monitoring of construction progress and generating building information modeling from unordered site image collections, Adv. Eng. Inform. 28 (1) (2014) 37–49, https://doi.org/10.1016/j.aei.2013.11.002.
[68] H. Kim, H. Kim, Y.W. Hong, H. Byun, Detecting construction equipment using a region-based fully convolutional network and transfer learning, J. Comput. Civ. Eng. 32 (2) (2017) 04017082, , https://doi.org/10.1061/(asce)cp.1943-5487.0000731.
[69] S. Han, S. Lee, A vision-based motion capture and recognition framework for behavior-based safety management, Autom. Constr. 35 (2013) 131–141, https://doi.org/10.1016/j.autcon.2013.05.001.
[70] L.Y. Ding, W.L. Fang, H.B. Luo, P.E.D. Love, B.T. Zhong, X. Ouyang, A deep hybrid learning model to detect unsafe behavior: integrating convolution neural networks and long short-term memory, Autom. Constr. 86 (2018) 118–124, https://doi.org/10.1016/j.autcon.2017.11.002.
[71] H. Kim, K. Kim, H. Kim, Vision-based object-centric safety assessment using fuzzy inference: monitoring struck-by accidents with moving objects, J. Comput. Civ. Eng. 30 (4) (2016) 13, , https://doi.org/10.1061/(asce)cp.1943-5487.0000562.
[72] Y. Fang, Y.K. Cho, J. Chen, A framework for real-time pro-active safety assistance for mobile crane lifting operations, Autom. Constr. 72 (2016) 367–379, https://doi.org/10.1016/j.autcon.2016.08.025.
[73] M. Golparvar-Fard, F. Peña-Mora, A. Arboleda Carlos, S. Lee, Visualization of construction progress monitoring with 4D simulation model overlaid on time-lapsed photographs, J. Comput. Civ. Eng. 23 (6) (2009) 391–404, https://doi.org/10.1061/(asce)0887-3801(2009)23:6(391.
[74] M. Golparvar-Fard, F. Pena-Mora, S. Savarese, Integrated sequential as-built and as-planned representation with D(4)AR tools in support of decision-making tasks in the AEC/FM industry, J. Constr. Eng. Manag. 137 (12) (2011) 1099–1116, https://doi.org/10.1061/(asce)co.1943-7862.0000371.
[75] A. Peddi, L. Huan, Y. Bai, S. Kim, Development of human pose analyzing algorithms for the determination of construction productivity in real-time, Construction Research Congress (2009), https://doi.org/10.1061/41020(339)2.
[76] M. Golparvar-Fard, F. Peña-Mora, Application of visualization techniques for construction progress monitoring, Computing in Civil Engineering (2007), https://doi.org/10.1061/40937(261)27.
[77] X.M. Zhang, J.Y. Zhang, M. Ma, Z.Q. Chen, S.L. Yue, T.T. He, X.B. Xu, A high precision quality inspection system for steel bars based on machine vision, Sensors 18 (8) (2018) 20, https://doi.org/10.3390/s18082732.
[78] Z.L. Wang, H. Li, X.L. Zhang, Construction waste recycling robot for nails and screws: computer vision technology and neural network approach, Autom. Constr. 97 (2019) 220–228, https://doi.org/10.1016/j.autcon.2018.11.009.
[79] E. Menendez, J.G. Victores, R. Montero, S. Martinez, C. Balaguer, Tunnel structural inspection and assessment using an autonomous robotic system, Autom. Constr. 87 (2018) 117–126, https://doi.org/10.1016/j.autcon.2017.12.001.
[80] Y.T. Zhu, Z.X. Zhang, Y.F. Zhu, X. Huang, Q.W. Zhuang, Capturing the cracking characteristics of concrete lining during prototype tests of a special-shaped tunnel using 3D DIC photogrammetry, Eur. J. Environ. Civ. Eng. 22 (2018) s179–s199, https://doi.org/10.1080/19648189.2017.1379445.
[81] G. Morgenthal, N. Hallermann, J. Kersten, J. Taraben, P. Debus, M. Helmrich, V. Rodehorst, Framework for automated UAS-based structural condition assessment of bridges, Autom. Constr. 97 (2019) 77–95, https://doi.org/10.1016/j.autcon.2018.10.006.
[82] P. Prasanna, K. Dana, N. Gucunski, B. Basily, Computer-vision based crack detection and analysis, International Society for Optics and Photonics (2012), https://doi.org/10.1117/12.915384.
[83] S. German, I. Brilakis, R. DesRoches, Rapid entropy-based detection and properties measurement of concrete spalling with machine vision for post-earthquake safety assessments, Adv. Eng. Inform. 26 (4) (2012) 846–858, https://doi.org/10.1016/j.aei.2012.06.005.
[84] R. Zaurin, N. Catbas, Integration of computer imaging and sensor data for structural health monitoring of bridges, Smart Mater. Struct. (2009), https://doi.org/10.1088/0964-1726/19/1/015019.
[85] T.T. Um, F.M.J. Pfister, D. Pichler, S. Endo, M. Lang, S. Hirche, U. Fietzek, D. Kulić, Data augmentation of wearable sensor data for Parkinson's disease monitoring using convolutional neural networks, Proceedings of the 19th ACM International Conference on Multimodal Interaction, 2017, https://doi.org/10.1145/3136755.3136817.
[86] B.-J. Koops, The trouble with European data protection law, International Data Privacy Law 4 (4) (2014) 226–250, https://doi.org/10.1093/idpl/ipu023.
[87] G.S. Alder, Ethical issues in electronic performance monitoring: a consideration of deontological and teleological perspectives, J. Bus. Ethics 17 (7) (1998) 729–743 https://link.springer.com/article/10.1023/A:1005776615072.
[88] K. Ball, Workplace surveillance: An overview, Labor History 51 (1) (2010) 87–106, https://doi.org/10.1080/00236561003654776.
[89] C.V. Dung, L.D. Anh, Autonomous concrete crack detection using deep fully convolutional neural network, Autom. Constr. 99 (2019) 52–58, https://doi.org/10.1016/j.autcon.2018.11.028.
[90] H. Kwaśnicka, L.C. Jain, Bridging the Semantic Gap in Image and Video Analysis, Springer, 2018, https://doi.org/10.1007/978-3-319-73891-8.
[91] B. Zhong, H. Wu, H. Li, S. Sepasgozar, H. Luo, L. He, A scientometric analysis and critical review of construction related ontology research, Autom. Constr. 101 (2019) 17–31, https://doi.org/10.1016/j.autcon.2017.04.002.

[92] X. Xing, B. Zhong, H. Luo, H. Li, H. Wu, Ontology for safety risk identification in metro construction, Comput. Ind. 109 (2019) 14–30, https://doi.org/10.1016/j.compind.2019.04.001.
[93] C.J. Anumba, R.R. Issa, J. Pan, I. Mutis, Ontology-based information and knowledge management in construction, Constr. Innov. 8 (3) (2008) 218–239, https://doi.org/10.1108/14714170810888976.
[94] K. Schwabe, J. Teizer, M. König, Applying rule-based model-checking to construction site layout planning tasks, Autom. Constr. 97 (2019) 205–219, https://doi.org/10.1016/j.autcon.2018.10.012.
[95] J. Tang, L. Khoja, H. Heinimann, Characterisation of survivability resilience with dynamic stock interdependence in financial networks, Applied Network Science 3 (1) (2018) 23, , https://doi.org/10.1007/s41109-018-0086-z.
[96] H. Wang, C. Schmid, Action recognition with improved trajectories, IEEE International Conference on Computer Vision (2013) 3551–3558.